

Engine Model

Marine Propulsion Engines — Marine Engineering Notes

1. Introduction to Type of Engine

Marine propulsion engines are generally grouped by fuel type, working cycle, and mechanism of power generation. The main categories are internal combustion diesel engines, further split into 2-stroke and 4-stroke designs, and external combustion steam turbine engines. The choice of engine type depends on the vessel's size, speed requirements, fuel economy targets, and operating profile.

Classification Basis:

- By working cycle — 2-stroke or 4-stroke diesel engines.
- By combustion type — internal combustion (diesel) vs external combustion (steam turbine).
- By speed — slow-speed, medium-speed, or high-speed engines.
- By application — main propulsion engines vs auxiliary generator engines.

Selection Factors:

- Ship size, deadweight and required propulsive power.
- Fuel efficiency and operating cost over the vessel's life.
- Space and weight constraints in the engine room.
- Maintenance requirements and crew familiarity.

2. Marine Diesel Engine — 2 Stroke

A two-stroke diesel engine completes the power cycle — intake, compression, combustion, and exhaust — in a single revolution of the crankshaft (2 piston strokes). It is the dominant choice for large ocean-going merchant ships as the main propulsion engine.

- Runs at low speed, typically 60–200 RPM, directly coupled to the propeller (no gearbox needed).
- One power stroke per revolution, giving higher power output per cylinder.
- Uses scavenging (forced air) to expel exhaust gases and draw in fresh air.
- Generally more fuel-efficient at large sizes, and can run on heavy fuel oil (HFO).
- Long-stroke, crosshead-type design to handle high loads at low speed.
- Common use — main propulsion engines on container ships, bulk carriers and tankers.

3. Marine Diesel Engine — 4 Stroke

A four-stroke diesel engine completes the power cycle over two crankshaft revolutions (4 piston strokes: intake, compression, power, exhaust). It is widely used for auxiliary generators and as main propulsion on smaller or medium-sized vessels.

- Runs at medium to high speed, typically 400–1,200+ RPM.
- Requires a gearbox/reduction gear to match propeller speed when used for propulsion.
- One power stroke every two revolutions — lighter and more compact than an equivalent 2-stroke engine.
- Commonly used to drive auxiliary generators supplying the ship's electrical power.

- Common use — auxiliary generator engines on almost all ship types.
- Common use — main propulsion on smaller vessels, ferries, tugs and offshore support craft.

4. Marine Steam Turbine Engine

A steam turbine is an external combustion engine that converts the thermal energy of high-pressure steam, generated in a boiler, into rotational mechanical energy by directing steam onto turbine blades.

- Steam is produced separately in a boiler, so fuel is burned externally rather than inside the cylinder.
- Very smooth, low-vibration operation due to continuous rotary motion (no reciprocating pistons).
- Historically used on large tankers, LNG carriers, and naval vessels.
- Largely superseded by diesel engines on most merchant ships due to lower fuel efficiency.
- Still relevant on some LNG carriers, where boil-off gas can be used as boiler fuel.
- Common use — older-generation LNG carriers, large tankers, and certain naval vessels.

5. Quick Reference Summary

Engine Type	Working Principle / Key Notes
Introduction (General)	Classification by cycle, combustion type, speed and application
Diesel — 2 Stroke	1 power stroke per revolution; low speed (60–200 RPM); direct-coupled to propeller
Diesel — 4 Stroke	1 power stroke per 2 revolutions; medium/high speed; needs gearbox
Steam Turbine	External combustion; continuous rotary motion; smooth & low-vibration

Note: Engine type selection depends on vessel size, speed, fuel economy and operating profile, and should always follow the manufacturer's operating manual and the ship's Planned Maintenance System (PMS).